Name: ______________________ **Date:** ______________________

Collaborators: ______________________ **Lab Section:** ______________________

Postlab Exercises (20 points)

Submit the postlab to the TA at the end of the lab.

Question 3 **2 points**

Record your measurements of the Helmholtz coil diameter D and radius $R = D/2$ and compute an average value in Data Table 1.

Data table 1: *Measurements of Helmholtz coil radius.*

D_{inner} (cm)	D_{outer} (cm)	R_{inner} (cm)	R_{outer} (cm)	$(R_{\text{inner}} + R_{\text{outer}})/2$ (cm)

$R = \overline{R} =$ ______________ m

Next, using the average value of the Helmholtz coil radius R, use eq. (7) to recompute the coefficient in eq. (8), assuming $N = 130$:

$$B = \underbrace{\left[\left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 N}{\overline{R}}\right]}_{\text{coefficient (T} \cdot \text{A)}} I.$$

In the experiment, you measured the diameter of the electron beam d, which is related to the radius of curvature by $r = d/2$. The calculation of e/m_e uses the product rB, so write a formula for computing rB given the measured current I and beam diameter d:

$rB =$ ______________ $d \cdot I$ (m · T)

Question 4 **3 points**

Record your measurements from the first voltage setting (350 V). Compute the mean of rB, the standard deviation, and the uncertainty on the mean, and estimate e/m_e.

Data table 2: *Beam measurements using first voltage setting.*

$V =$ 350 V

N	Coil current I (A)	Beam diameter d (m)	rB (m · T)
1			
2			
3			
4			
5			
6			
7			
8			
9			

Mean $\overline{rB}$:

Standard deviation σ_{rB}:

Uncertainty on the mean $\sigma_{\overline{rB}} = \sigma_{rB}/\sqrt{9}$:

Estimated e/m_e:

Uncertainty on e/m_e:

Final result: $e/m_e \pm$ uncertainty, with units:

Question 5 **3 points**

Record your measurements from the second voltage setting (400 V). Compute the mean of rB, the standard deviation, and the uncertainty on the mean, and estimate e/m_e.

Data table 3: *Beam measurements using second voltage setting.*

$V =$ 400 V

N	Coil current I (A)	Beam diameter d (m)	rB (m · T)
1			
2			
3			
4			
5			
6			
7			
8			
9			

Mean $\overline{rB}$:

Standard deviation σ_{rB}:

Uncertainty on the mean $\sigma_{\overline{rB}} = \sigma_{rB}/\sqrt{9}$:

Estimated e/m_e:

Uncertainty on e/m_e:

Final result: $e/m_e \pm$ uncertainty, with units:

Question 6 **3 points**

Record your measurements from the third voltage setting. Compute the mean of rB, the standard deviation, and the uncertainty on the mean, and estimate e/m_e.

Data table 4: *Beam measurements using third voltage setting.*

$V =$ ____________________ V

N	Coil current I (A)	Beam diameter d (m)	rB (m · T)
1			
2			
3			
4			
5			
6			
7			
8			
9			

Mean $\overline{rB}$:

Standard deviation σ_{rB}:

Uncertainty on the mean $\sigma_{\overline{rB}} = \sigma_{rB}/\sqrt{9}$:

Estimated e/m_e:

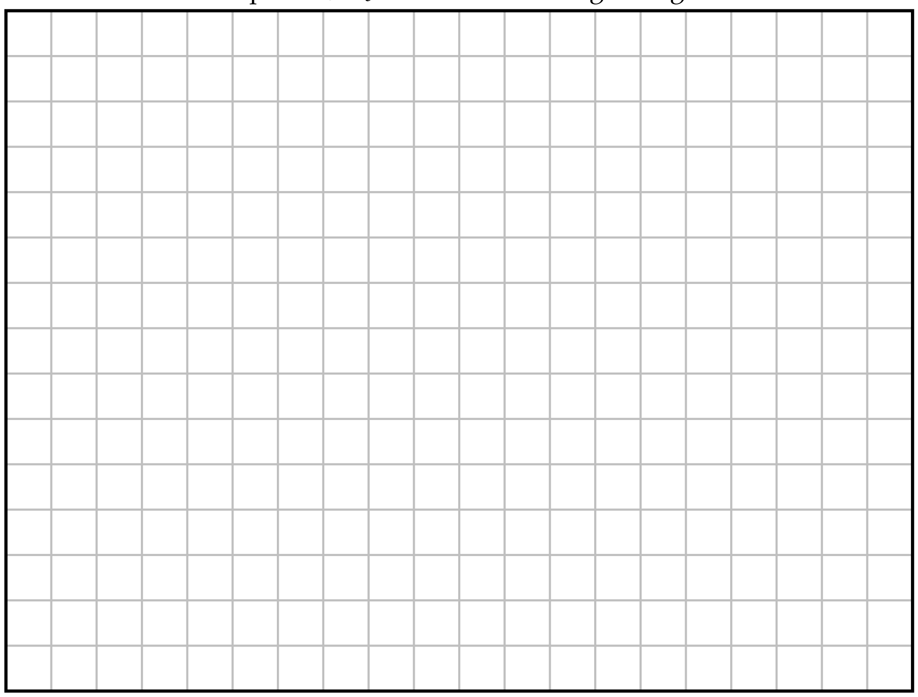
Uncertainty on e/m_e:

Final result: $e/m_e \pm$ uncertainty, with units:

Question 7 **4 points**

In Graph 1, plot e/m_e vs. accelerating voltage V and include error bars determined from your calculation of the uncertainty in e/m_e. Include axes labels with units. Use scientific notation in your y-axis label by stating it in terms of 10^{11} C/kg. Draw a horizontal line at the y-axis value of $e/m_e = 1.759 \times 10^{11}$ C/kg, the accepted value of e/m_e.

Graph 1: e/m_e *versus accelerating voltage.*



Question 8 **3 points**

Taking into account each measured value with its associated error, are your results consistent with each other? With the true value? Explain your answer and discuss why or why not.

Question 9 **2 points**

Name at least two of the major sources of uncertainty in the measurement of e/m_e in this lab. Be specific (e.g., “human error” as a reason is vague and imprecise).